

Linearity

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What are these notes?

These are notes for a first course in Linear Algebra.

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Preface

This section discusses why these notes exist. Students may skip this section.

Why publish a new set of Linear Algebra notes?

Linear Algebra, like Calculus, is one of the math subjects with the most textbooks, so it's reasonable to ask why a new set of notes is needed. Plainly, I looked at the six open-access books on the subject on the AIMath website and found that none of them were fit for my purpose (detailed below).

The perspective of these notes

These notes are constructed to vindicate the following objectives:

1. It is morally right for course materials to be free. Few existing books on this subject in English satisfy this criterion (eg Hefferon (2022) is GFDL or CC BY-SA 3.0 US, Treil (2024) is CC BY-NC-ND 3.0 with source unavailable). This book, and the source used to generate it are freely available with a permissive license.
2. In practice, scientists, engineers, etc. need to be able to recognize linearity so that they may choose the correct solution techniques. They also need to understand why linear problems are preferable to non-linear ones so that they might try to massage their current problem into a linear one.
3. When we say “solution techniques” as above, 99% of the time we mean software packages. Mathematicians and physicists teach linear algebra techniques in colleges and universities, and emphasize by-hand solution techniques for historical and cultural reasons. Most working people who encounter such problems do not use such techniques, they recognize that their problem is linear and offload the problem to a software package. Mathematicians and physicists generally get a second pass at learning linear algebra in a more theory-heavy context (at the very least when learning modules), and so do not need that approach in a first course.

4. The usefulness of linear algebra techniques stems wholly from the homomorphism property of linear maps:

$$L(aV + bW) = aL(V) + bL(W)$$

No introductory, open-access, English language books on the topic that I am aware of motivate the study of the subject with this point. They traditionally begin with coordinate geometry or solving systems of linear equations. It is a very mathematicians' way of thinking to motivate study of a topic by identifying a class of equations and asking "How do we solve them? What properties do they have?" This is not a way of thinking that is useful for people encountering linearity in the wild.

Many advanced undergraduate books like Axler (2024) begin with the definition of a vector space.

It is also a very mathematician's way of thinking to begin with a definition of a set of objects which will be mapped into or out of (viz. most treatments of Abstract Algebra). To talk about maps (which are really the objects of interest), *surely* we must first talk about (co)domains!

Coordinate geometry is, at least, a class of real problems where linear techniques naturally arise, but the relevance of this as an example from "the wild" has basically vanished in the last 70 years. Today's scientists and engineers are more likely to encounter linearity in optimization, data science, machine learning, or numerical PDEs.

So, for these reasons, I set out to write my own course notes.

Part I

Vector Spaces

Chapter 1

What is Linearity?

The function $C : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$C(r) := 2\pi r$$

computes the circumference of a circle, given its radius.

Notation

First, let's talk about this notation.

The notation $C : \mathbb{R} \rightarrow \mathbb{R}$ tells you about the inputs and outputs of the function C . When we write $f : A \rightarrow B$, we mean that f takes inputs from the set A and creates outputs in the set B . You can think of f as a machine transforming A s into B s.

The notation $C(r) := 2\pi r$ tells you that the *definition* of the function C appears here. This is to avoid confusion like the following:

$$f(x) = x^2$$

If the function f hasn't appeared before, then this equation is probably a definition. But if we wrote "Given $f(x) = x - 2$, solve the equation $f(x) = x^2$ ", then the same equation is not a definition. To avoid this ambiguity, when we write an equals sign with a colon on one side like this $A := B$ or $B =: A$, we mean that the name on the side of the $=$ *with* the colon is defined to be the expression on the side without.

The circumference of a circle has a couple nice properties. First, the circumference of a circle of radius 14 is twice the circumference of a circle of radius 7:

$$\begin{aligned}C(7) &= 2\pi(7) = 14\pi \\C(14) &= 2\pi(14) = 28\pi\end{aligned}$$

This holds in general; multiplying the radius of a circle by k also changes the circumference by a factor of k :

$$C(kr) = kC(r).$$

Furthermore, adding any amount to the radius increases the circumference in a predictable way:

$$C(r_1 + r_2) = C(r_1) + C(r_2).$$

It's a bit remarkable that these two properties hold not just for circles; scaling any shape in the plane (with circumference c) by a factor of k multiplies its circumference by k (its new circumference is kc), and increasing the scale by a constant s increases its circumference by sc .

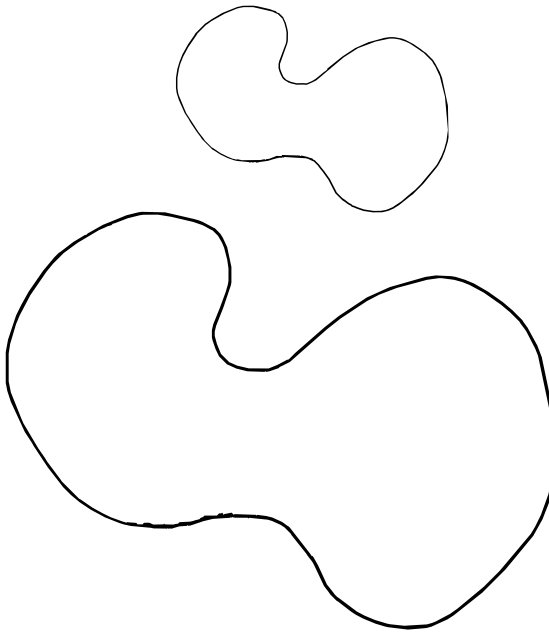


Figure 1.1: Scaling a curve by a factor of 2

1.1 Two properties

What other functions have the following two properties?

Property 1: $f(ax) = af(x)$

Property 2: $f(x + y) = f(x) + f(y)$

Exercise 1.1.

Demonstrate that the *area* of a circle as a function of its radius does not satisfy properties 1 and 2.

Exercise 1.2.

Can you think of any shape in the plane whose area (as a function of scaling) satisfies properties 1 and 2? If yes, which? If no, why not?

Exercise 1.3.

Can you think of any shape in the 3-dimensional space whose *volume* (as a function of scaling) satisfies properties 1 and 2? If yes, which? If no, why not?

Classification of functions satisfying Properties 1 and 2

So far, we have seen that some functions satisfy properties 1 and 2, and others do not.

Table 1.1: Functions that do and don't satisfy properties 1 and 2

do	don't
circumference of a circle as a function of radius	area of a circle as a function of radius
circumference of any shape in the plane as a function of scale	area of any shape in the plane as a function of scale
	volume of any shape in 3-dimensions as a function of scale

Why does scaling satisfy properties 1 & 2 for any shape, not just circles?

We asserted above that property 1 is not just satisfied by circles (when you scale the radius) but is satisfied by all curves. Why is this the case?

What do we mean by scaling a figure in the plane by a factor of 2? Well, a reasonable answer is to say that a figure is a set of points and each point has an x and y coordinate. For example, the circle of radius r is the set of points with coordinates given by

$$\begin{aligned}x &= r \cos \theta \\ y &= r \sin \theta\end{aligned}$$

where $\theta \in [0, 2\pi]$.

Notation

We introduced a symbol above: $\in$. This literally means “in” or “is in” depending on context.

For example, $\pi \in \mathbb{R}$ means “pi is in the real numbers”.

Sometimes we will want to list the set first, like $\mathbb{R} \ni \pi$. In this case we read it as “the real numbers contain pi.”

Notation

When we want to refer to a set, we will often use notation like the following:

$$\left\{ (x, y) \in \mathbb{R}^2 \left| \begin{array}{l} x = r \cos \theta \\ y = r \sin \theta \end{array} \right. , \theta \in [0, 2\pi] \right\}.$$

In general, this notation has the form

$$B = \left\{ f(x_1, x_2, \dots) \in A \left| \begin{array}{l} \text{constraints involving} \\ x_1, x_2, \dots \end{array} \right. \right\}.$$

The process for constructing the set B is the following:

1. Find all the x_i that satisfy the constraints to the right of the $|$ symbol.
2. Plug all the x_i you found in the previous step into the function f .
3. The function f produces things in A , and so the set of all the things you produced in the last step is a collection of some (but not necessarily all) of the things in A .
This is the set B .

Consider the point $(4, 2)$ in the plane. We can think of this point as the “sum” of its x and y coordinates:

$$(4, 2) = (4, 0) + (0, 2).$$

To scale this point by a factor of 2, it seems reasonable to multiply both coordinates by 2:

$$S_2((4, 2)) = (8, 4).$$

Notice that the function $S_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ “multiply coordinates by 2” has properties 1 and 2.

Exercise 1.4.

Verify this.

Exercise 1.5.

Check that applying S_2 to a circle of radius r produces a circle of radius $2r$.

Now, for any curve $\gamma : [a, b] \rightarrow \mathbb{R}^2$ given by

$$\gamma(t) = (x(t), y(t)),$$

its length can be computed by

$$\int_a^b \sqrt{x'(t)^2 + y'(t)^2} \, dt.$$

(You may have seen this in multivariable calculus or physics.) Consider the composition

$$[a, b] \xrightarrow{\gamma} \mathbb{R}^2 \xrightarrow{S_2} \mathbb{R}^2$$

The formula is

$$(S_2 \circ \gamma)(t) = (2x(t), 2y(t))$$

and the length is given by

$$\begin{aligned} & \int_a^b \sqrt{([2x(t)])'^2 + ([2y(t)])'^2} \, dt \\ &= \int_a^b \sqrt{(2x'(t))^2 + (2y'(t))^2} \, dt \\ &= \int_a^b \sqrt{4(x'(t))^2 + 4(y'(t))^2} \, dt \\ &= \int_a^b 2\sqrt{(x'(t))^2 + (y'(t))^2} \, dt \\ &= 2 \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} \, dt \end{aligned}$$

Notice that what we obtain on the last line is exactly twice the length of the curve γ . Convince yourself that there is nothing special about the number 2 here; if we had replaced S_2 by S_{17} , then we would have obtained 17 times the length of γ in the last line.

More functions which break up over “sums”

Differentiation

Now consider how we compute the derivative of a function like the following:

$$\begin{aligned}\frac{d}{dx} [2x^2 + x] &= \frac{d}{dx} [2x^2] + \frac{d}{dx} [x] \\ &= 2 \frac{d}{dx} [x^2] + \frac{d}{dx} [x] \\ &= 4x + 1\end{aligned}$$

If we let $D : \text{FUNCTIONS} \rightarrow \text{FUNCTIONS}$ be the operation of taking a derivative, then in the first line we used

$$D[f_1(x) + f_2(x)] = D[f_1(x)] + D[f_2(x)]$$

and in the second line we used the fact that, when k is constant,

$$D[kf(x)] = kD[f(x)].$$

Thus, D (that is, differentiation) satisfies properties 1 and 2. (Although you should probably be uncomfortable that we wrote $D : \text{FUNCTIONS} \rightarrow \text{FUNCTIONS}$ above. What is the set FUNCTIONS ? Are all functions differentiable? We will address this later. For now, it suffices to replace FUNCTIONS above with P_n , the set of all polynomials of degree at most n . In fact, it is the case that $D : P_n \rightarrow P_{n-1}$ if $n \geq 1$.)

Definite integration

Fix an interval $[a, b]$ and consider

$$I(f) := \int_a^b f(x) \, dx.$$

Exercise 1.6.

Check that I has properties 1 and 2.

Why are properties 1 and 2 useful?

Why is it useful that D satisfies properties 1 and 2? It allows us to compute derivatives of complicated expressions like $2x^2 + x$ if we only know the computation on some simple parts of the expression. Knowing the derivative of x^2 and x is all that is needed.

Similarly, if we know the circumference of a shape in the plane at one scale, we can compute its circumference at all scales using property 1.

Not all functions are linear, but if a function is linear, it is much easier to compute with.

1.2 Linear functions

Definition 1.1 (linear function). A function satisfying properties 1 and 2 is called *linear*.

 Warning

We use the term “linear” for these functions, but we also use the word “line” for graphs in the plane with formula $f(x) = mx + b$. This term is overloaded and means different things in these two contexts.

Exercise 1.7.

Show that $f(x) = mx + b$ is only a linear function when $b = 0$.

Chapter 2

The notion of a Vector Space

If $L : A \rightarrow B$ is linear, what must be true about A and B ?

Let's go back to the definition of a linear function. A function is linear if and only if it satisfies the following two properties:

$$\text{Property 1: } L(ax) = aL(x)$$

$$\text{Property 2: } L(x + y) = L(x) + L(y).$$

Let's list a few things that must be true to arrive at these expressions:

- there are terms (like a in property 1) that we can factor through L
- there are terms (like x, y in properties 1 and 2) that we cannot factor through L
- there is some kind of addition on the x s and y s

There are perhaps some more properties that would be nice, and that are true about all the domains and codomains of linear functions we have seen so far:

- the a s (that we can pull through L) have nice algebraic properties ($+$, $-$, $\times$, division)
- the $+$ operation on the x s and y s has some nice properties, too (existence of an identity, commutativity, etc)

Eventually, mathematicians (who were working with linear functions intuitively) worked out the minimal set of facts that one needs about the domain and codomain of a linear function for everything to be coherent. Here it is

2.1 Definition

Definition 2.1. A *vector space* is a set V of *vectors* and a set F of *scalars* that satisfy the following properties

- the set of scalars F is a field (see the next section for more information on what we mean by field),
- there is a function $+: V \times V \rightarrow V$, called *vector addition* that
 - is associative: $(X + Y) + Z = X + (Y + Z)$,
 - is commutative: $X + Y = Y + X$,
 - has a 0 (an additive identity) which means that 0 satisfies $0 + X = X$ for all $X \in V$,
 - has negatives (additive inverses) which means that for each $X \in V$ there is an element $Y \in V$ such that $X + Y = 0$ (one can prove that there is only one inverse of X , and this is usually written $-X$),
- there is a function $\cdot: F \times V \rightarrow V$, called *scalar multiplication* such that
 - scalar multiplication associates with field multiplication (that is, $\forall a, b \in F, X \in V$ it's true that $(ab)X = a(bX)$)
 - the multiplicative unit in F is a unit for scalar multiplication (that is, $\forall X \in V$ we have $1X = X$)
- the following distributive laws hold:
 - $(a + b)X = aX + bX, \quad \forall a, b \in F, X \in V$
 - $a(X + Y) = aX + aY, \quad \forall a \in F, X, Y \in V$



Notation

Above, we used the symbol $\forall$. This is a symbol that mathematicians use that literally just means “for all”.

So, as an example $x \in P_3, \forall x \in P_2$ is read as “for all x in the set of second degree polynomials, x is in the set of third degree polynomials”, which is just a wordy way of saying that all second degree polynomials are third degree polynomials.

Note that the notation $+: V \times V \rightarrow V$ implies that V is, in particular, closed under addition (since V is the codomain of $+$; addition cannot by definition return an element not in V). Similarly we can conclude that V must be closed under scalar multiplication.

All of the properties above hold (with the field of scalars F set to $\mathbb{R}$) for all the spaces we have used so far as the domain and codomain of a linear function. (You may want to check this for yourself quickly. Most of these facts obviously hold for polynomials, functions, points in $\mathbb{R}^2$, and $\mathbb{R}$ itself. However, one must still check all of them before using a set as the domain or codomain of a linear function.)

Not all sets of mathematical objects satisfy these properties, though. An example is the set of points on the sphere. It turns out that there is no way (although

the proof will have to wait for a future chapter) to turn that set into a vector space.

The way you should think about this definition is the following:

- we are **allowed to use** a set A satisfying all of the properties in Definition 2.1 as the domain or codomain of a linear function, but
- a set A that fails to satisfy any of the properties above **can never be used** as the domain or codomain of a linear function. If you tried to use it in this way, you would eventually run into statements that make no sense.¹

An example of the kind of problem you can run into is the following: Let $L : A \rightarrow B$ be linear and suppose A is a vector space as defined above but that B fails to have an additive identity (that is, there is no zero vector in B). We can compute

$$L(0) = 0L(0) = 0(?) = 0$$

where the $?$ stands for whatever $L(0)$ maps to. Notice that in the end, it doesn't matter because we conclude that $L(0) = 0 \in B$. But this is nonsensical since we assumed there was no zero vector in B .

Exercise 2.1. Check that $\mathbb{R}$ forms a vector space (the set of scalars is $\mathbb{R}$).

Exercise 2.2. Check that $\mathbb{R}^2$ is a vector space.

2.2 Scalars

In the above definition, we said that the scalars for a vector space must come from a field. What does this mean?

For the purposes of this text, our field will always be $\mathbb{R}$ or the field of complex numbers:

$$\mathbb{C} := \{a + bi \mid a, b \in \mathbb{R}\}$$

where the symbol i has the property that $i^2 = -1$ (and so we sometimes write it as $\sqrt{-1}$). These two sets are the set of scalars for most vector spaces found in applications in the wild.

Exercise 2.3. Check that $\mathbb{C}$ can be thought of as a vector space with field of scalars $\mathbb{R}$.

¹It is the case that some maps between spaces that are weaker than vector spaces have defining properties 1 and 2. The spaces are a generalization of vector spaces called modules and the maps are called module homomorphisms. If you look at the definition of module homomorphism on wikipedia, you'll see the same two equations we used to define linear functions.

Definition 2.2. Let P_n be the set (which we mentioned informally in the previous chapter)

$$\{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \mid a_n, a_{n-1}, \dots, a_1, a_0 \in \mathbb{R}\}.$$

This is called the *set of polynomials of degree n* .

Exercise 2.4. Is P_n a vector space for each $n \in \mathbb{N}$?

From the previous two examples, you might notice that the definitions

$$\mathbb{C} := \{a + bi \mid a, b \in \mathbb{R}\}$$

and

$$P_n := \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \mid a_n, a_{n-1}, \dots, a_1, a_0 \in \mathbb{R}\}$$

look similar, and they give you a hint about how to think of these as vector spaces; in both cases, they are constructed as the sum of things with coefficients in $\mathbb{R}$. We will investigate this further in the next chapter.

Fields

In case you are interested, the full definition of a field can be found on Wikipedia. A field that you already know about, but that is not $\mathbb{R}$ or $\mathbb{C}$ is the field of rational numbers:

$$\mathbb{Q} := \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}.$$

There are esoteric examples of fields, and also algebraic structures stranger than fields. If this is interesting to you, you might try to take a course in Abstract Algebra. I learned this topic from Liz Stanhope and Dummit and Foote (2004).

In computing, real numbers (which do form a field) are often represented by 32 or 64 bit floating point numbers. It is perhaps interesting to know that floating point numbers (in any number of bits) (which are encountered **often** in the wild) do not form a field because addition of floating point numbers is not associative.

2.3 Subspaces

We have seen in this chapter that if you want to do something with a linear function, you better check first that its domain and codomain are vector spaces. It's quite a lengthy process to verify all the properties in Definition 2.1, even if in many cases it is not difficult to verify each one individually. We are in need

of machines that allow us to know that something is a vector space more easily. That is what this and the next section are about.

How do we know that a set $W \subset V$ is a vector space?

Notation

Above, we used the symbol $\subset$. This just means “is a sub set of”, for example $\mathbb{Z} \subset \mathbb{Q}$ means “the set of integers is a subset of the set of rational numbers.”

When we say “A is a subset of B” or, equivalently write $A \subset B$, we mean that everything in A is also in B .

We will sometimes write $A \supset B$ to mean that B is a subset of A or that “A has B as a subset.”

We’ve already seen some examples of this. For example, we know that differentiable functions form a vector space, and also the set P_n forms a vector space. Since every polynomial is differentiable, it follows that $P_n \subset \{\text{differentiable functions}\}$. So P_n is an example of a subset of a vector space that turns out to be a vector space itself.

Let’s introduce some terminology for this situation.

Definition 2.3. A subset W of a vector space V is called a (vector) *subspace* if W is itself a vector space (with field of scalars F , addition, multiplication, zero vector inherited from V).

We have already seen an example of a subspace above. A nonexample is the set of points in the plane at distance 1 from the origin. This set doesn’t satisfy the closure assumptions of Definition 2.1.

It is a bit remarkable (although not difficult to prove) that we actually only need to check the closure assumptions. The following theorem encodes this fact.

Theorem 2.1. *A subset W of a vector space V is itself a vector space (with field of scalars F , addition, multiplication, zero vector inherited from V) iff the following two properties hold:*

1. *W is closed under scalar multiplication: $aX \in W \quad \forall a \in F, X \in W$*
2. *W is closed under vector addition: $X + Y \in W \quad \forall X, Y \in W$*

We won’t do the proof here (you can go to e.g. Hefferon (2022) to see it), but we will sketch it.

Since the scalars, and the operations $\cdot, +$ are inherited from the ambient space V , they are known to satisfy associativity, commutativity, existence of inverses and zeros, etc (because we had to prove those things when we showed that V was a vector space). However, in Definition 2.1, we wrote $+: V \times V \rightarrow V$. That is, $+$ takes two vectors in V and returns a vector in V . Since $W \subset V$, we know that we can use elements of W as inputs to $+$. That is, we know $+: W \times W \rightarrow V$,

but we cannot know addition of vectors always lands in the subset W . The first assumption above tells us this. Similarly for scalar multiplication.

For the proof in the opposite direction, notice that for W to be a vector space we must have $+: W \times W \rightarrow W$. If addition is not closed, then $+$ has the wrong codomain, and W isn't a vector space. Similarly if W is not closed under scalar multiplication.

Easier subspaces

Notice that we can use Theorem 2.1 to quite easily prove (assuming we know differentiable functions form a vector space) that P_n is a vector space. We can also conclude easily that, for example,

$$\{a \sin x + b \cos x \mid a, b \in \mathbb{R}\}$$

is a vector space.

Exercise 2.5. Is the set of arrows that are either fully horizontal or fully vertical a vector subspace of the vector space of arrows in 2D? Why?

Exercise 2.6. Consider the set of monomials

$$\{1, x, x^2, x^3, \dots\}.$$

If I pick any two of these (say x^k and x^l with $k \neq l$, eg x^{17} and x^{2025}) and form the set

$$\{ax^k + bx^l \mid a, b \in \mathbb{R}\},$$

is this a vector subspace of the space of differentiable functions? Why?

2.4 The space of lists of length n

We have seen that the following are vector spaces: $\mathbb{R}$ over $\mathbb{R}$, $\mathbb{R}^2$ over $\mathbb{R}$, $\mathbb{C}$ over $\mathbb{C}$.

There is a very powerful theorem that we can prove to generate many vector spaces that are of a similar form.

Theorem 2.2. *Let F be any field. Define the following set*

$$F^n := \{(a_1, a_2, \dots, a_n) \mid a_i \in F\}.$$

This is the space of lists of length n , with members from F .

Let $+$: $F^n \times F^n \rightarrow F^n$ be defined by

$$(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n) = (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

and let $\cdot$: $F \times F^n \rightarrow F^n$ be defined by

$$x(a_1, a_2, \dots, a_n) = (xa_1, xa_2, \dots, xa_n).$$

Then F^n with these two operations (and field of scalars F) is a vector space.

The proof is a straightforward check of all the items in Definition 2.1.

Note that $\mathbb{R}, \mathbb{R}^2, \mathbb{C}$ can be proved to be vector spaces using this theorem.

To develop your intuition for what this looks like, let's do a multiplication in $\mathbb{C}^4$:

$$\begin{aligned} & i(1, i, 1 + i, 2 - 3i) \\ &= (i, i^2, i + i^2, 2i - 3i^2) \\ &= (i, -1, i - 1, 2i + 3). \end{aligned}$$

Note that this theorem only works if F^n is a vector space over F . So, above, $\mathbb{C}^4$ is a vector space with scalars $\mathbb{C}$.

Exercise 2.7.

Is $\mathbb{R}^n$ a vector subspace of $\mathbb{C}^n$? Think very carefully about Definition 2.3.

Chapter 3

Bases

Remember that the derivative function $D(f) := \frac{df}{dx}$ allows us to compute the derivative of, for example

$$f(x) = x^2 + 4 \sin x$$

if we know how to compute D on only x^2 and $\sin x$. In fact, knowing just how to compute D on these two functions and knowing that D is linear allows us to compute the derivative of any function of the form

$$x \mapsto ax^2 + b \sin x, \quad a, b \in \mathbb{R}.$$

Notation

We just introduced the notation $\mapsto$. This is a mathematician's way of referring to a function without giving it a name.

As an example both of

$$x \mapsto x^2 \quad \text{and} \quad f(x) = x^2$$

are the same function. On the right, we choose to give the function a name “ f ”. On the left, we describe the function by saying how inputs are transformed into outputs, but we don't give it a name. The function on the left is just “the function that squares its input”. It doesn't have a name, so this is called an *anonymous function*.

Mathematicians use an anonymous function when they won't want to refer to a function again later, and so have no need of giving it a name. If you'll want to refer to a function later, it's useful to give it a name when you define it, so you'll write $f(x) = x^2$ (or use some other letter if f is already

used for something else).

This pattern also shows up in computer science, although they often call anonymous functions *lambdas* and instead of writing $x \mapsto x^2$ they will write `x.x^2` or `fun x => x^2` or some variation thereof.

From a very small amount of information, we actually know a lot about D . Thus, we begin this chapter straightforwardly with a question, which is the title of the next section:

3.1 What is the minimal amount of information needed to unambiguously describe a linear function?

Consider the following example:

Assume that $L : P_2 \rightarrow P_0$ is linear and that

$$L(x^2 + 3x) = 4.$$

Can we determine L on every element of P_2 ? That is, can we compute

$$L(ax^2 + bx + c)?$$

Let's try our best. We would like to isolate a term that looks like $x^2 + 3x$ because that is something we have information about.

$$\begin{aligned} L(ax^2 + bx + c) &= L(ax^2 + bx) + L(c) && \text{by property 1} \\ &= L\left(a\left[x^2 + \frac{b}{a}x\right]\right) + L(c) && \text{assuming } a \neq 0 \\ &= aL\left(x^2 + \frac{b}{a}x\right) + L(c) && \text{by property 2} \end{aligned}$$

then we are basically stuck. Although we can force x^2 to show up, we can't *at the same time* force $3x$ to show up; forcing a coefficient of 1 on x^2 will always effect the coefficient of x . Furthermore, we have no information about how to deal with $L(c)$.

So, in this case, knowing that L is linear and knowing its value on *one* input is insufficient.

Exercise 3.1. In the example above, is it enough to know that

$$\begin{aligned} L(x^2) &= 1, \\ L(x) &= 1, \end{aligned}$$

to be able to determine L on all of P_2 ?

What if we know

$$\begin{aligned} L(x^2) &= 1, \\ L(x) &= 1, \quad ? \\ L(1) &= -2 \end{aligned}$$

3.2 Writing vectors relative to a set of vectors

Let $X, Y \in \mathbb{R}^2$ be given by $X := (1, 1)$ and $Y := (1, -1)$. Suppose $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ is linear and also that we know

$$\begin{aligned} L(X) &= 1, \\ L(Y) &= -2. \end{aligned}$$

Can we compute L on the vector $(2, 3)$? If the answer to this question were yes then there would have to be a way to express $(2, 3)$ in terms of X and Y (since those are the only two values we know anything about). That is, there would be $a, b \in \mathbb{R}$ such that

$$(2, 3) = aX + bY = a(1, 1) + b(1, -1) = (a, a) + (b, -b) = (a + b, a - b).$$

So, L can be computed on $(2, 3)$ if and only if the following system has a solution:

$$\begin{aligned} 2 &= a + b, \\ 3 &= a - b. \end{aligned}$$

We can solve this system (solving for, say, a in the first and substituting the value in the second). We find that there is only one solution: $a = \frac{5}{2}, b = -\frac{1}{2}$.

Using this information, how do we compute $L(2, 3)$? Well, now we know that

$$(2, 3) = \frac{5}{2}(1, 1) + \left(-\frac{1}{2}\right)(1, -1)$$

so let's just apply the function L to both sides of this equation:

$$\begin{aligned}
L(2, 3) &= L \left[\frac{5}{2}(1, 1) + \left(-\frac{1}{2}\right)(1, -1) \right] \\
&= L \left[\frac{5}{2}(1, 1) \right] + L \left[\left(-\frac{1}{2}\right)(1, -1) \right] \\
&= \frac{5}{2}L(1, 1) + \left(-\frac{1}{2}\right)L(1, -1) \\
&= \frac{5}{2}1 + \left(-\frac{1}{2}\right)(-2) \\
&= \frac{5}{2} + 1 \\
&= \frac{7}{2}
\end{aligned}$$

Exercise 3.2. For each $=$ -symbol in the computation above, write the assumption, property, or rule that tells us we are allowed to conclude the left hand side is equal to the right hand side.

The above computation shows that the answer to our question is yes, this amount of information about L is sufficient to compute $L(2, 3)$. Notice that this is the case even though we were not given the formula for L .

You may want to convince yourself that the vector $(2, 3)$ above was not special; given the information we have about L , we can compute $L(k, l)$ for any $(k, l) \in \mathbb{R}^2$.

Exercise 3.3. If we know that $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ is linear and that

$$L(0, 1) = 1, \quad L(0, 2) = 2,$$

can we compute $L(1, 2)$? If yes, compute it. If no, why not?

Exercise 3.4. If we know that $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ is linear and that

$$L(0, 1) = 2, \quad L(1, 1) = 2,$$

can we compute $L(1, 2)$? If yes, compute it. If no, why not?

Linear combinations

The trick to computing $L(2, 3)$ in the previous section was to rewrite $(2, 3)$ in the form

$$(2, 3) = a(1, 1) + b(1, -1).$$

This is a good ansatz¹ because we are able to split over the $+$ and pull the a and b out (using the fact that L is linear). The following definition captures this pattern in general.

Definition 3.1. If $X_1, X_2, \dots, X_n$ is a collection of vectors in the vector space V , then any expression of the form

$$a_1X_1 + a_2X_2 + \dots + a_nX_n, \quad a_i \in F$$

is called a *linear combination* of $X_1, X_2, \dots, X_n$.

For some of the following exercises, we need to know about the following definition:

Definition 3.2. We use $C^k(\mathbb{R}, \mathbb{R})$ to denote the set of functions $\mathbb{R} \rightarrow \mathbb{R}$ such that, $\forall f \in C^k(\mathbb{R}, \mathbb{R})$

- f has at least k derivatives and
- all of f 's k derivatives are continuous.

Sometimes we write just C^k rather than $C^k(\mathbb{R}, \mathbb{R})$ if it is implicit from the context that we are talking about functions $\mathbb{R} \rightarrow \mathbb{R}$.

Notice, in particular, that C^0 is just the set of continuous functions $\mathbb{R} \rightarrow \mathbb{R}$. C^1 is the set of functions with at least one derivative which is continuous.

C^∞ is used to denote the set of functions $\mathbb{R} \rightarrow \mathbb{R}$ that have infinitely many derivatives, all of which are continuous (many “nice” familiar functions are in C^∞ , for example $x \mapsto \sin x$, $x \mapsto e^x$ and any polynomial).

Lemma 3.1. For any k , the set $C^k(\mathbb{R}, \mathbb{R})$ is a vector space over $\mathbb{R}$.

The proof of this lemma follows from the fact that $\mathbb{R}$ is a vector space over $\mathbb{R}$, and when we do operations on functions, we really do operations inside the codomain of the function.

Exercise 3.5. Is

$$\pi \sin x - 100x^3$$

a linear combination in C^0 ? If no, why not? If yes, what is a set of vectors in C^0 that this is a linear combination of?

¹This is a German word that, in math, means a guess with some free parameters (in this case, a and b) in it. We compute with this guess, and then later find values of the parameters that make the computation valid and also solve our problem. If we cannot find such values, then we have essentially proved that there is no solution to our problem with the form given in the ansatz (although there may be a solution of another form if our ansatz is not general enough).

Exercise 3.6. Is

$$7 \cos x \sin x$$

a linear combination in C^0 ? If no, why not? If yes, what is a set of vectors in C^0 that this is a linear combination of?

We then have the following generalization of the computation we did in the example above:

Theorem 3.1. *Let $L : A \rightarrow B$ be linear. These two statements are equivalent:*

- *We know the value of L on a set of vectors $X_1, X_2, \dots, X_n \in A$. That is,*

$$\begin{aligned} L(X_1) &= Y_1 \\ L(X_2) &= Y_2 \\ &\vdots \\ L(X_n) &= Y_n \end{aligned}$$

for some $Y_i \in B$.

- *We know the value of L on any linear combination of $X_1, X_2, \dots, X_n$.*

The proof $\Rightarrow$ is a straightforward application of properties 1 and 2 of linear functions. You might try to generate the justification yourself. The proof of $\Leftarrow$ is an application of proof by contradiction.

i Note

Theorem 3.1 is not a tautology!

The set $\{X_1, X_2, \dots, X_n\}$ is a finite set (think of $\{(1, 1), (1, -1)\}$ which contains only 2 elements).

The set of linear combinations of $\{X_1, X_2, \dots, X_n\}$ is *infinite* (if the field of scalars is infinite, which in the case of $\mathbb{R}$ or $\mathbb{C}$ it is).

The set of linear combinations of $\{(1, 1), (1, -1)\}$ contains, for example,

$$\begin{aligned} (1, 1), (2, 2), (a, a) \forall a \in \mathbb{R}, \\ (\pi, e), (\sqrt{2}, 2025^{1/3}), \\ a(1, 1) + b(1, -1) = (a + b, a - b) \forall a, b \in \mathbb{R}. \end{aligned}$$

In particular, this set is infinite.

The span of a set of vectors

We will use the words “is a linear combination of” or “can be expressed as a linear combination of” quite often, so we introduce some terminology to shorthand

this.

Definition 3.3. If V is a vector space over a field F , $X_1, \dots, X_n \in V$, and W is expressible as a linear combination of $\{X_1, \dots, X_n\}$ we say that W is in the *span* of $\{X_1, \dots, X_n\}$. We can write this

$$W \in \text{Span}\{X_1, \dots, X_n\}.$$

We also refer to the following set as *the span* of $\{X_1, \dots, X_n\}$:

$$\text{Span}\{X_1, \dots, X_n\} := \{a_1 X_1 + \dots + a_n X_n \mid a_i \in F\}.$$

So, for example, $(0, 2025) \in \text{Span}\{(0, 1), (0, 6)\}$, but the vector $(17, 9) \notin \text{Span}\{(0, 1), (0, 6)\}$.

Exercise 3.7. Let $(a, b) \in \mathbb{R}^2$. Is $(a, b) \in \text{Span}\{(1, 1), (1, -1)\}$? Why or why not?

Exercise 3.8. Let $(a, b) \in \mathbb{R}^2$. Is $(a, b) \in \text{Span}\{(1, 2), (1, 5), (1, -3)\}$? Why or why not?

Given the above definition, we can rephrase Theorem 3.1:

Theorem 3.2. Let $L : A \rightarrow B$ be linear. These two statements are equivalent:

- We know the value of L on a set of vectors $X_1, X_2, \dots, X_n \in A$.
- We know the value of L on $\text{Span}\{X_1, X_2, \dots, X_n\}$.

There isn't really anything to prove here; this is exactly Theorem 3.1 except with "any linear combination of" replaced by *Span*, which is Definition 3.3.

Exercise 3.9. Is $\text{Span}\{1 + x, x + x^2\} = P_2$? Why or why not?

Exercise 3.10. Show that $\text{Span}\{1 + x, x + x^2, x^2, x^3\} = P_3$ by finding a representation of

$$ax^3 + bx^2 + cx + d$$

(which is the form of a generic element of P_3) as a linear combination of $\{1 + x, x + x^2, x^2, x^3\}$.

Exercise 3.11. Suppose we know that $F'(x) = f(x)$ and $G'(x) = g(x)$ (that is, F, G have derivatives, and we give the names f, g to those derivatives). Recall the definition of the operator D for differentiation:

$$D(h) := h'(x),$$

and that we already know $D : \{\text{differentiable functions}\} \rightarrow \{\text{functions}\}$ is linear. (In view of the notation introduced in Definition 3.2, we can now write this as $D : C^1 \rightarrow C^0$.)

Is it true that

$$D[F(x) + 14G(x)] = f(x) + 14g(x)?$$

Did you use any properties about D *other than* linearity to conclude that?

Suppose $G(x) > 0, \forall x$. Is it true that

$$D \left[\frac{F(x)}{G(x)} \right] = \frac{f(x)G(x) - F(x)g(x)}{[G(x)]^2}?$$

Did you use any properties about D *other than* linearity to conclude that?

Linear independence

Suppose we have a collection of vectors $\{X_1, \dots, X_n\} \in \mathbb{R}^2$. We have seen examples where

1. $\{X_1, \dots, X_n\}$ consists of two vectors and $\text{Span}\{X_1, \dots, X_n\}$ **is not all of** $\mathbb{R}^2$
(this happened when $\{X_1, \dots, X_n\} = \{(0, 1), (0, 2)\}$),
2. $\{X_1, \dots, X_n\}$ consists of two vectors and $\text{Span}\{X_1, \dots, X_n\}$ **is equal to** $\mathbb{R}^2$
(this happened when $\{X_1, \dots, X_n\} = \{(1, 1), (1, -1)\}$),
3. $\{X_1, \dots, X_n\}$ consists of three vectors and $\text{Span}\{X_1, \dots, X_n\}$ **is equal to** $\mathbb{R}^2$
(this happened when $\{X_1, \dots, X_n\} = \{(1, 2), (1, 6), (1, -3)\}$).

In the last case, we got something extra; there were infinitely many ways of representing an arbitrary element $(a, b) \in \mathbb{R}^2$ as a linear combination of $\{(1, 2), (1, 6), (1, -3)\}$. In the second case, we get a unique way of representing each vector as a linear combination, and in the first case we don't even get any way to represent some vectors as a linear combination.

From Definition 3.3 we now can say that the distinction between the first case and the other two is that

$$\text{Span}\{(1, 1), (1, -1)\} = \text{Span}\{(1, 2), (1, 6), (1, -3)\} = \mathbb{R}^2$$

but that

$$\text{Span}\{(0, 1), (0, 2)\} \neq \mathbb{R}^2.$$

From Theorem 3.2, we know that if we know a linear function on the two vectors

$$(1, 1), (1, -1)$$

or on the three vectors

$$(1, 2), (1, 6), (1, -3)$$

then we know its value on all inputs $(a, b) \in \mathbb{R}^2$.

What is the criterion that distinguishes the second case from the last one? That is, is there a property we can measure about the set of vectors

$$\{(1, 2), (1, 6), (1, -3)\}$$

that is different when we measure it about

$$\{(1, 1), (1, -1)\}$$

that tells is that we get a unique linear combination in the latter case, but a nonunique linear combination in the former case? It turns out to be the following definition, and we will see why in a later part of the book.

Definition 3.4. Let V be a vector space and $\{X_1, \dots, X_n\} \subset V$. We say that the set $\{X_1, \dots, X_n\}$ is *linearly independent* if, for each $i \in \{1, \dots, n\}$,

$$X_i \notin \text{Span}\{X_1, X_2, \dots, X_{i-1}, X_{i+1}, \dots, X_n\}.$$

This is another way of saying that X_i is **not** in the span of all the $\{X_j\}$ with X_i removed.

Exercise 3.12. Is the list $[(1, 0), (0, 1)] \subset \mathbb{R}^2$ linearly independent? Explain why.

Exercise 3.13. Is the list $[(2, 1), (2, -1)] \subset \mathbb{R}^2$ linearly independent? Explain why.

Exercise 3.14. Is the list $[(1, 2), (3, 4), (-1, 4)] \subset \mathbb{R}^2$ linearly independent? Explain why.

Although we chose to define linear independence using Definition 3.4, there are equivalent statements that we could have used. We summarize them in the following theorem.

Theorem 3.3. Let V be a vector space and $\{X_1, \dots, X_n\} \subset V$. The following statements are equivalent:

1. $\{X_1, \dots, X_n\}$ is linearly independent in the sense of Definition 3.4.
2. $0 = a_1 X_1 + \dots + a_n X_n$ has only one solution (when $a_i = 0 \forall i$).
3. For every $v \in \text{Span}\{X_1, \dots, X_n\}$, the equation $v = a_1 X_1 + \dots + a_n X_n$ has **exactly** one solution.

When we say “the following statements are equivalent”, we mean that among (1.), (2.), and (3.) above, there are proofs

$$(1.) \Leftrightarrow (2.) \Leftrightarrow (3.).$$

So, we chose to use Definition 3.4 as the definition, and (in the proof of Theorem 3.3) would prove that (2.) is equivalent to Definition 3.4. But we could have just as well used (2.) as the definition, and proved the statement in Definition 3.4 as a theorem. If you look at different books on this subject, you will see different approaches; all of them are equally correct.

We omit the proof of Theorem 3.3; the equivalences are not difficult to prove.

From this, we get the following lemma quite easily:

Lemma 3.2. *Let V be a vector space and $S = \{X_1, \dots, X_n\} \subset V$.*

- *If S contains the zero vector, it is linearly dependent.*
- *If S contains the same vector twice, it is linearly dependent.*

Note that this lemma **is not an iff** statement. For example, not every linearly dependent set contains the zero vector. But if a set does contain the zero vector, this lemma tells you that that set is linearly dependent.

Necessity and sufficiency

We are finally in a position to answer the question that began this chapter:

What is the minimal amount of information needed to unambiguously describe a linear function?

Note that there are two parts of this statement: how much information suffices to define a linear function, and how much information minimally suffices. These criteria are different.

For example, in Chapter 1 we saw that the formula

$$C(r) := 2\pi r$$

is enough to know C on all its inputs (after all, this formula tells you how to compute C for any input you could pick). So this formula **suffices** to define the linear function C .

However, we saw later in that chapter that, once we know C is linear, this is more information than we need. It is enough to know far less information. For example, suppose I tell you that, in some other universe there are creatures called pumans who have a shape they call a pircle whose pircumference (as they call it) is a function of some measurement they refer to as the pradius of the pircle. Both pradius and pircumference are nonnegative real numbers, thankfully.

I don't know the formula for pircumference as a function of pradius (communication with the pumans is very slow and difficult), but I have learned from them two things:

- pircumference as a function of pradius is linear
- the pircumference of a pircle of pradius 2025 is 45.

It turns out that this is enough information to discover the formula for pircumference! Let $k \geq 0$ be any arbitrary pradius, and call the pircumference function P . We can compute

$$P(k) = P\left(\frac{k}{2025}2025\right) = \frac{k}{2025}P(2025) = \frac{k}{2025}45 = \frac{k}{45}.$$

Verify for yourself that you understand why each equality in this computation is true. For the second equality, we used our knowledge that pircumference is linear and for the third equality we used our knowledge that $P(2025) = 45$. If we had been missing either of those facts, we could not conclusively determine the formula for P . That means that, once we know P is linear, the knowledge of P on at least one nonzero input is **necessary**.

The following theorem gives two criteria addressing our motivating question at the start of this section: the first criterion describes when we have sufficient information to determine a linear function, the second describes when we have only the necessary (minimal) amount of information.

Theorem 3.4. *Let $L : A \rightarrow B$ be linear. Suppose we know the value of L on a set of vectors $\{V_1, \dots, V_n\}$.*

- *If $\text{Span}\{V_1, \dots, V_n\} = A$, we can compute the value of L on all inputs.*
- *If $\{V_1, \dots, V_n\}$ is linearly independent, then $\{V_1, \dots, V_n\}$ is a minimal set of information defining L on $\text{Span}\{V_1, \dots, V_n\}$.*

Note that, if $\text{Span}\{V_1, \dots, V_n\} = A$ and $\{V_1, \dots, V_n\}$ is linearly independent, this theorem implies that $\{V_1, \dots, V_n\}$ is a minimal choice of information defining L everywhere on A .

This theorem is very powerful. For example, consider a linear function $F : \mathbb{R}^3 \rightarrow \mathbb{R}^{2025}$. Theorem 3.4 tells us that if we know only three facts, e.g. the values of $L(1, 0, 0)$, $L(0, 1, 0)$, and $L(0, 0, 1)$, then we are able to use this information to find the value of L on **any** of the infinitely many vectors in $\mathbb{R}^3$. Of course, we would first have to show that $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ spans $\mathbb{R}^3$.

The situation of having a set of vectors that both spans an entire vector space and is linearly independent is so common, we give it the following name:

A basis for a vector space

Definition 3.5. Let V be a vector space and let B be a finite, ordered list of vectors in V . If

- $\text{Span } B = V$ and
- B is linearly independent,

we say that B is a *basis* for V .

We write list here because order and repetition matter, and these are not qualities of sets.

Notation

We write lists like $[1, 2, 2, 3]$ with square brackets rather than curly ones.² Note that for lists, $[1, 1, 1] \neq [1]$ while for sets $\{1, 1, 1\} = \{1\}$, and also, for lists $[1, 2] \neq [2, 1]$ while for sets $\{1, 2\} = \{2, 1\}$.

Exercise 3.15. Is the list $[(1, 0), (0, 1)] \subset \mathbb{R}^2$ a basis? Explain why.

Exercise 3.16. Is the set $[(2, 1), (2, -1)] \subset \mathbb{R}^2$ a basis? Explain why.

Note

The preceding exercises demonstrate a very important point: if V has a basis, it is not unique. There are actually many bases for a vector space, and which one you should choose to use **depends on the problem at hand**.

For example, both $[(1, 0), (0, 1)]$ and $[(1, 1), (1, -1)]$ are bases for $\mathbb{R}^2$. If we are dealing with a linear function $L : \mathbb{R}^2 \rightarrow \mathbb{R}$ where we know $L(1, 1) = 5$ and $L(1, -1) = 3$, then it is not useful to work with the basis $[(1, 0), (0, 1)]$.

We now have a restatement of Theorem 3.4 that uses this definition:

Theorem 3.5. *Let $L : A \rightarrow B$ be linear. Suppose we know the value of L on a set of vectors $\{V_1, \dots, V_n\}$. If $[V_1, \dots, V_n]$ is a basis for A , then we know the value of L on all its inputs and $\{V_1, \dots, V_n\}$ is a minimal choice of data defining L .*

We finally answered the two mysteries that have been motivating us for a few chapters, so we're done, right?

²This is the notation convention used by many modern programming languages.

Chapter 4

Vector spaces isomorphic to Euclidean space

No, there are still more mysteries!

One example is the difference between P_1 and $\mathbb{R}^2$. We have the following ordered bases for these

$$[x, 1] \text{ for } P_1 \text{ and } [(1, 0), (0, 1)] \text{ for } \mathbb{R}^2.$$

When we have fixed a basis for a vector space, Definition 3.5 together with Theorem 3.3 and Definition 3.3 tell us that, for each vector in our vector space, there is a exactly one way to write that vector as a linear combination of the basis vectors. So, for example, the vector $x - 3 \in P_1$ can be thought of as

$$(1) \underbrace{\begin{pmatrix} x \\ 1 \end{pmatrix}}_{\text{1st basis vector}} + (-3) \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\text{2nd basis vector}}.$$

If you and I are thinking of the same basis $[x, 1]$ for P_1 , then there is no confusion if we just list the coefficients of this vector relative to this basis. That is, we use the notation $(1, -3)$ to *refer* to the vector $x - 3 \in P_1$.

But $(1, -3)$ is a list of two real numbers, so isn't this an element in $\mathbb{R}^2$? Is P_1 the same as $\mathbb{R}^2$?

4.1 Dimension

One remarkable fact about P_1 is that, although $[x, 1]$ isn't the only basis for it (the list $[x + 1, x - 1]$, we have seen, is another), **all** the bases of P_1 have something in common: they have exactly two elements. This can be generalized to the following lemma:

Lemma 4.1. *Let V be a vector space with at least one finite basis and let B be that basis. Then all bases of V have the same number of elements as B .¹*

This immediately tells us that all bases for P_1 have two elements, as well as all bases for $\mathbb{R}^2$.

Since all bases for a vector space are the same length we can refer to this length as a property of the vector space, rather than a property of a particular basis.

Definition 4.1. Let V be a vector space with a finite basis of size n . Then we say that the *dimension* of V is n and we write this like

$$\dim V = n.$$

So, using this notation, $\dim P_1 = 2 = \dim \mathbb{R}^2$.

Exercise 4.1.

What is the dimension of P_2 ? What about P_3 ? What about P_n ?

Exercise 4.2.

What is the dimension of the vector space of arrows drawn on a piece of paper? What is the dimension of the vector space of arrows created in a space like a room? (You can think of bits of string of different lengths, with arrows on one end of them, held in space by two people. Addition and multiplication is as for arrows on paper.)

Exercise 4.3.

What is the dimension of $\mathbb{C}$ as a vector space over itself?

Exercise 4.4.

What is the dimension of $\mathbb{C}$ as a vector space over $\mathbb{R}$?

The following lemma says that if one vector space is inside another, then the dimension of the one on the inside can't be larger than the one on the outside:

Lemma 4.2. *Let $W \subset V$ be a vector subspace of the vector space V . Suppose W had finite dimension. Then*

$$\dim W \leq \dim V.$$

Exercise 4.5.

The dimension of the space of continuous functions $\mathbb{R} \rightarrow \mathbb{R}$ is not finite. Use the result of Exercise 4.1 to explain why.

Exercise 4.6.

¹For readers familiar with the notion of cardinality, this lemma can be generalized to infinite bases where it states that any two bases have the same cardinality.

Recall that we know that the function $D : C^1 \rightarrow C^0$ “take the derivative” is linear. Let K be the set of all vectors in C^1 which are taken to the 0 vector in C^0 by D . That is, K is the set of all differentiable functions whose first derivative is 0 everywhere. As an equation,

$$K := \{f : \mathbb{R} \rightarrow \mathbb{R} \mid D(f) = f'(x) = 0\}.$$

K is a vector subspace of C^1 (you might want to convince yourself that this is true using Theorem 2.1, but you don’t have to for this exercise). What is $\dim K$?

4.2 The notion of isomorphism

We have identified one property that P_1 and $\mathbb{R}^2$ have in common: they have the same dimension. There is a more general notion of “sameness” that is true for these two vector spaces. We will introduce the word for it here, before explaining precisely what it means in the following section.

For the purposes of this book, V and W are *isomorphic* if they are “the same” once we fix some piece of data.

In mathematics more generally, if two objects are isomorphic, it means that they “behave identically” but might need an extra piece of information to actually identify one with the other. The full notion is extremely broad.²

4.3 Isomorphism to $\mathbb{R}^n$ (given a basis)

Recall Theorem 2.2, that $\mathbb{R}^n$ is the set of lists of length n , where the elements of the list are real numbers. Recall that $\mathbb{R}^n$ is a vector space, and notice that we can demonstrate at least one basis for $\mathbb{R}^n$:

$$\begin{aligned} &(1, 0, 0, \dots, 0, 0), \\ &(0, 1, 0, \dots, 0, 0), \\ &\vdots \\ &(0, 0, \dots, 0, 1, 0), \\ &(0, 0, 0, \dots, 0, 1) \end{aligned}$$

Notice that this basis has exactly n elements.

Now, suppose we have another vector space V over $\mathbb{R}$ with a basis of n elements $[b_1, \dots, b_n]$. Given a vector $v \in V$, there is a unique list of n real numbers $v_1, \dots, v_n$ such that

$$v = v_1 b_1 + v_2 b_2 + \dots + v_n b_n.$$

²<https://en.wikipedia.org/wiki/Isomorphism>

As an example, consider P_1 with basis $[x + 1, x - 1]$. The vector $12\pi x - 2025$ can only be written in one way relative to this basis:

$$12\pi x - 2025 = \left(\frac{12\pi - 2025}{2}\right)(x + 1) + \left(\frac{12\pi + 2025}{2}\right)(x - 1).$$

Once we choose a basis, every vector can be “identified” with its list of coefficients relative to that basis. In the example above, we have

$$12\pi x - 2025 \quad " = " \quad \left(\frac{12\pi - 2025}{2}, \frac{12\pi + 2025}{2}\right).$$

We put the equals sign in quotes here because these two vectors are not identical (they’re not even in the same vector space), but once we choose a basis, we can think of the left as the right.

Exercise 4.7.

Choose a different basis for P_1 and write the coordinates of the vector $12\pi x - 2025$ relative to that basis.

Exercise 4.8.

Choose any basis you want for P_2 .

The set $W := \text{Span}\{(1, 0, 0, 0), (0, 0, 1, 0), (0, 0, 0, 1), (1, 0, 1, 0)\} \subset \mathbb{R}^4$ is a vector subspace (But you shouldn’t believe me! You can check for yourself!). Choose a basis for this subspace.

Using your chosen bases, what vector is $(1, 2, 3) \in \mathbb{R}^3$ when interpreted as coordinates for a vector in P_2 ? How about when interpreted as coordinates for a vector in W ?

To make this notion precise, we notice that there is a function (which depends on the basis we choose for P_1) which we’ll name $C : P_1 \rightarrow \mathbb{R}^2$. It turns out that this function is a linear function between vector spaces.

In particular, adding in P_1 and converting to coordinates, adding, and converting back to a polynomial produce the same value:

$$(ax + b) + (cx + d) = (a + c)x + (b + d)$$

and

$$\begin{aligned} & \underbrace{\left(\frac{a+b}{2}, \frac{a-b}{2}\right)}_{C(ax+b)} + \underbrace{\left(\frac{c+d}{2}, \frac{c-d}{2}\right)}_{C(cx+d)} = \left(\frac{a+b+c+d}{2}, \frac{a+c-(b+d)}{2}\right) \\ C^{-1} \left(\frac{a+b+c+d}{2}, \frac{a+c-(b+d)}{2}\right) &= \frac{a+b+c+d}{2}(x+1) + \frac{a+c-(b+d)}{2}(x-1) \\ &= (a+c)x + (b+d) \end{aligned}$$

where C^{-1} is the inverse function.

A similar property holds for scalar multiplication.

It is perhaps clear that these properties are not unique to P_1 and $\mathbb{R}^2$, nor to the basis we chose for P_1 in the above example. Generalizing this example, we arrive at the following theorem.

Theorem 4.1. *Let V be a finite dimensional vector space over the field F . Let $n = \dim V$, and let B be a choice of basis for V . Then the function $V \rightarrow F^n$ taking a vector to its list of coordinates relative to B is a linear function which is injective and surjective. (That is, the function is an isomorphism of vector spaces.) The inverse of this function is also linear.*

The fact that the coordinate function $\Phi : V \rightarrow F^n$ and its inverse are linear imply that addition and scalar multiplication in V and F^n are compatible. That is, we can add vectors in V , or we can convert to coordinates, add in F^n and convert back to V and get the same answer.

$$\begin{array}{ccc} V \times V & \xrightarrow{+} & V \\ \Phi \downarrow & & \downarrow \Phi \\ F^n \times F^n & \xrightarrow{+} & F^n \end{array}$$

Note that the $+$ symbol on the top of the diagram is the vector addition operation in V . The $+$ symbol on the bottom is vector addition in F^n .

The same rule applies for scalar multiplication.

$$\begin{array}{ccc} F \times V & \xrightarrow{\cdot} & V \\ \Phi \downarrow & & \downarrow \Phi \\ F \times F^n & \xrightarrow{\cdot} & F^n \end{array}$$

Note that the $\cdot$ symbol on the top of the diagram is the scalar multiplication operation in V . The $\cdot$ symbol on the bottom is scalar multiplication in F^n .

Exercise 4.9.

Consider two bases $B_1 := [x^2, x, 1]$ and $B_2 := [x^2, x, -1]$ for P_2 .

Consider the vector $(17, 5, 2) \in \mathbb{R}^3$. What vector in P_2 does this represent relative to B_1 ? What about relative to B_2 ?

What Theorem 4.1 says is that, once we choose a basis for P_1 , we can treat it *the same as* $\mathbb{R}^2$. We can work with polynomials or coordinates in $\mathbb{R}^2$ and we will get the same results. The same thing is true for all vector spaces over $\mathbb{R}$ with dimension 2. This fact is generalized in the following corollary:

Corollary 4.1. *Let V and W be two vector spaces over the same field of scalars F . If $\dim V$ and $\dim W$ are finite and equal, then V and W are “the same” (isomorphic) once we choose bases for each of them.*

The proof of this fact is that, since they have the same dimension (call it n), then both V and W can be thought of as lists of coordinates in F^n .

Exercise 4.10.

Consider

$$v = 3x^3 + 5x^2 - 7x + 1, w = -4x^3 + 2x^2 - 4x + 7$$

in the vector space P_3 .

1. Compute the polynomial $v + w$.
2. Choose a basis for P_3 . Write v and w as lists of coordinates in this basis.
3. Add these two lists of coordinates using addition as it is defined in $\mathbb{R}^4$.
4. Write $v + w$ (your answer in part 1) as a list of coordinates in your basis for P_3 .
5. Did you get the same answer in the previous two parts? If the answer is no, go back and check your work.

Exercise 4.11.

Consider

$$v = 2x^2 - 2x + 7$$

in the vector space P_2 .

1. Compute the polynomial $5v$.
2. Choose a basis for P_2 . Write v as a list of coordinates in this basis.
3. Multiply this list of coordinates by 5 using scalar multiplication as it is defined in $\mathbb{R}^3$.
4. Write $5v$ (your answer in part 1) as a list of coordinates in your basis for P_2 .
5. Did you get the same answer in the previous two parts? If the answer is no, go back and check your work.

Chapter 5

Ways of writing vectors (relative to a basis)

We saw in Theorem 4.1 that, once we choose a basis, we may think of a vector as its list of coefficients relative to that basis. Let's look at that in more detail.

5.1 Linear combinations

Consider the vector space P_3 over $\mathbb{R}$. We know that $[1, x, x^2, x^3]$ is a basis for P_3 . When we want to write a vector, say

$$2x^3 + x^2 - \pi$$

as a list of coefficients relative to this basis, how do we do that?

We always write the vector as a linear combination of basis vectors:

$$2x^3 + x^2 - \pi = (2)x^3 + (1)x^2 + (0)x + (-\pi)1.$$

Here, the numbers in parentheses are the coefficients of this vector (polynomial) relative to this basis.

Be careful! We had to remember to write a coefficient for **every** basis vector, even if that vector doesn't appear on the left side. So we must include some scalar times x , even though there are no x s on the left. How do we do that? We use a coefficient of zero (note that, no matter what field we are using for scalars, we always have a coefficient of “zero”; it's the additive identity in the field).

Be careful again! Had we considered a different basis, we would get a different list of coefficients in our linear combination. Refer back to Exercise 4.9 to see an example of this.

5.2 Columns

Once we choose a basis, it is common to write the coordinates of a that vector as a column, rather than in the list notation we have used so far. So, for example, the vector $2x^3 + x^2 - \pi \in P_3$ above would be written as

$$\begin{bmatrix} 2 \\ 1 \\ 0 \\ -\pi \end{bmatrix}.$$

Lets again remember that this notation only works if we have already agreed on a basis for P_3 (in this case, $[x^3, x^2, x, 1]$).

You may have seen notation like

$$\begin{pmatrix} 2 \\ 1 \\ 0 \\ -\pi \end{pmatrix}.$$

This is merely a syntactical difference; both of these notations are perfectly reasonable. In this book we will use square brackets, rather than round ones.

5.3 Other ways you might encounter

There are other notations that are used in some communities. This is because the concept of a vector space has been re-discovered many times. The first that we will talk about is common in physics.

i, j, k

When we are considering the vector space $\mathbb{R}^3$, it is common to use the letters i, j, k to mean the following three vectors:

$$\begin{aligned} i &= (1, 0, 0) \\ j &= (0, 1, 0) \\ k &= (0, 0, 1) \end{aligned}$$

Sometimes, to visually distinguish these on the printed page, they are written in bold (eg **i, j, k**), or with hats above them (eg $\hat{i}, \hat{j}, \hat{k}$). Many other notations are out there in the wild for these three vectors in $\mathbb{R}^3$.

So, for example, the vector we would have written $(1, 2, 0)$ up until now would be written as $i + 2j$. This has the convenience that we can write nothing for

basis elements with a coefficient of zero. It also has the advantage that it is a commutative notation; the vector $i + 2j$ is the same as the vector $2j + i$.

Beware, though, that this notation almost always applies to a single situation: when we are talking about the vector space $\mathbb{R}^3$ and we have already agreed that the basis is $[(1, 0, 0), (0, 1, 0), (0, 0, 1)]$. This is a common situation in Physics (specifically classical mechanics), and that is the reason this notation has become standard there.

This notation is sometimes also used for $\mathbb{R}^2$, where $i = (1, 0)$ and $j = (0, 1)$. This makes implicit use of the fact that there is a linear map $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by

$$(x, y) \mapsto (x, y, 0)$$

so that $-i + 6j \in \mathbb{R}^2$ (which we would usually write as $(-1, 6)$) corresponds to $-i + 6j \in \mathbb{R}^3$ (which is really $(-1, 6, 0)$). This map is a bijection between $\mathbb{R}^2$ and $\text{Span}\{(1, 0, 0), (0, 1, 0)\} \subset \mathbb{R}^3$. As long as this is understood by the people communicating about vectors, all is good.

But beware that the function $(x, y) \mapsto (x, y, 0)$ is not canonical! There are many injective linear functions $\mathbb{R}^2 \rightarrow \mathbb{R}^3$ and none of them is preferred over another (at least, mathematically). For example, $(x, y) \mapsto (x + y, 0, x - y)$ is also an injective linear function, but it does not send $i \in \mathbb{R}^2$ to $i \in \mathbb{R}^3$.

Rows

It is rare (in mathematics) these days¹, but some texts choose to represent the coefficients of a vector as a row, rather than a column. So, for example, they would represent $2x^3 + x^2 - \pi \in P_3$ as

$$[2 \ 1 \ 0 \ -\pi] \quad \text{or} \quad (2 \ 1 \ 0 \ -\pi) \quad \text{or} \quad (2, 1, 0, -\pi).$$

We have used the last of these three notations in Theorem 2.2. Writing vectors in this way has implications for how one must then write linear functions in coordinates (more on this later). In this text, when we intend to combine a vector in coordinates with a linear function in coordinates, we write the vector vertically.

bra-ket

In Quantum Mechanics, it is common to work with vectors that are functions. If we are in a finite dimensional vector space of functions (eg P_n), then we can write vectors as lists of coordinates (once we choose a basis). But when we are working with an infinite dimensional vector space (such as C^0 or C^∞), we have

¹It is common to see horizontally written vectors in programming languages. For example, the invocation `np.array([1, 2, 3])` in the NumPy library for the Python programming language.

no hope of writing vectors as lists of coefficients relative to a basis (because the list may be infinitely long).

In these cases, it is common to simply write the name of the function (eg $f \in C^\infty$) and then refer elsewhere for information about the properties of the function.

In Quantum Mechanics, it is common to write the name of a function inside a $\langle$ and a $|$ to emphasize that it is a vector:

$$\langle f |.$$

So, for example, if we know that $\langle f |$ and $\langle g |$ are vectors in the same vector space, then $\langle f + 3g |$ is also a vector in the same space.

Summation notation

In the study of manifolds², including differentiable geometry and general relativity, it is common to write the basis for an n -dimensional vector space like

$$\left[\frac{d}{dx^1}, \frac{d}{dx^2}, \dots, \frac{d}{dx^n} \right]$$

where $\frac{d}{dx^i}$ is just a symbolic placeholder for the i -th basis vector (akin to how $\hat{j}$ is a symbolic placeholder for the second basis vector in a particular basis of $\mathbb{R}^3$).

In this case, we would write a vector like

$$\frac{d}{dx^1} + 2\frac{d}{dx^2} - \pi\frac{d}{dx^3}.$$

When we want to talk about a vector in this notation, but without reference to particular coefficients, it's common to write

$$X^1 \frac{d}{dx^1} + X^2 \frac{d}{dx^2} + \dots + X^n \frac{d}{dx^n}.$$

Note that the superscript on X^1 is not an exponent; it's just a way of notationally distinguishing a symbol. Think of it akin to the notation $a_1, a_2, \dots$. So, for example, we might write the following:

The length of the vector

$$X^1 \frac{d}{dx^1} + X^2 \frac{d}{dx^2} + \dots + X^n \frac{d}{dx^n}$$

is

$$\sqrt{(X^1)^2 + (X^2)^2 + \dots + (X^n)^2}.$$

²<https://en.wikipedia.org/wiki/Manifold>

Implicit sums

When working with vectors we write like

$$X^1 \frac{d}{dx^1} + X^2 \frac{d}{dx^2} + \cdots + X^n \frac{d}{dx^n}$$

it is common to write them using a capital Σ like

$$\sum_{i=1}^n X^i \frac{d}{dx^i}.$$

You might remember this notation from discussions of Riemann sums in Calculus.

A stranger thing is common in the study of manifolds; if the smallest and largest values of i are understood, it is common to omit the Σ altogether. This situation occurs quite often; the lower limit of i is 1 and the upper limit is just the dimension of the vector space in which $X^1 \frac{d}{dx^1} + X^2 \frac{d}{dx^2} + \cdots + X^n \frac{d}{dx^n}$ lives.

So, all three of the following notations *denote* the same vector:³

$$\begin{aligned} & X^1 \frac{d}{dx^1} + X^2 \frac{d}{dx^2} + \cdots + X^n \frac{d}{dx^n} \\ &= \sum_{i=1}^n X^i \frac{d}{dx^i} \\ &= X^i \frac{d}{dx^i} \end{aligned}.$$

³The last notation of the three in the following equation is called the Einstein summation convention

Chapter 6

Ways of writing linear maps

We saw in the previous chapter that, once we choose a basis for a vector space V , we can write vectors as their list of coefficients relative to this basis.

Suppose we have a linear function $L : V \rightarrow W$. How can we write this in a basis?

6.1 Maps in finite dimensions

Let's consider the linear function $F : P_2 \rightarrow \mathbb{R}^2$ given by

$$F(x^2) = (0, 0), \quad F(x) = (1, 0), \quad F(1) = (0, 1).$$

Let's express this in a basis. Consider the basis $[x^2, x, 1]$ for P_2 and $[(1, 0), (0, 1)]$ for $\mathbb{R}^2$.

Then the vector $x \in P_2$ corresponds to the list of coefficients

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

and the vector $(1, 0) \in \mathbb{R}^2$ corresponds to the list of coefficients

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

So, whatever our expression for F is in coordinates, it must fit in

$$F \left(\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix},$$

and similarly,

$$F \left(\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad F \left(\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

First, let's note that, when we're applying a linear function to the coefficients of a vector written as a column, we usually don't write the parentheses after the function name. That is, we will write function applications like

$$F \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad F \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad F \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Writing linear functions as 2-dimensional arrays.

We will introduce the notation for writing F above in coordinates. Then we will develop fast processes for computing function application, composition, sums, multiples, and inverses of linear functions all while remaining “in coordinates”.

The motivation will be the following:

If we know a linear function on a basis for its domain, we know it everywhere. In particular,

$$\begin{aligned} F \begin{bmatrix} a \\ b \\ c \end{bmatrix} &= F \left(a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right) \\ &= aF \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + bF \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + cF \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\ &= a \begin{bmatrix} 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} (0)a + (1)b + (0)c \\ (0)a + (0)b + (1)c \end{bmatrix}. \end{aligned}$$

The way we will write F in coordinates, is to write it as a 2-dimensional array of numbers.

$$\begin{bmatrix} - & - & - \\ - & - & - \end{bmatrix}.$$

The number of columns of the array is the dimension of the **domain** of F , so in this case there are three columns. The number of rows of the array is the dimension of the **codomain** of F , so in this case there are two rows.

The first column of the array is where F sends the first basis vector of its domain. If we use the basis we have already chosen, then the array for F looks like

$$\begin{bmatrix} 0 & - & - \\ 0 & - & - \end{bmatrix}$$

because F takes its first basis vector to $(0, 0)$, which, in the coordinates for the codomain, is the column

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Similarly, the second column of the array is where F sends the second basis vector

$$\begin{bmatrix} 0 & 1 & - \\ 0 & 0 & - \end{bmatrix}$$

and the third column is where F sends the third basis vector

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

This two dimensional array of numbers we used to represent F is called a *matrix*.

Exercise 6.1.

Consider the linear function $G : P_2 \rightarrow \mathbb{R}^2$ given by

$$G(x^2) = (2, 0), \quad G(x) = (-1, 3), \quad G(1) = (7, \pi).$$

Using the same basis that we used for F above, write the matrix for G .

The way we apply a matrix, is as we did in the example above

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a_{11}x + a_{12}y + a_{13}z \\ a_{21}x + a_{22}y + a_{23}z \end{bmatrix}.$$

The formula for a matrix with n rows and m columns is needlessly complex. The intuition is this:

- The input vector must have the same number of entries as the number of columns of the matrix.
- The output vector must have the same number of entries as the number of rows of the matrix.
- To get the **first** entry in the output, multiply each of the entries in the input column with each of the entries in the **first** row of the matrix. Then add the products. That is the **first** entry of the result.

- To get the **second** entry in the output, multiply each of the entries in the input column with each of the entries in the **second** row of the matrix. Then add the products. That is the **second** entry of the result.
- Continue in this manner until you have filled all the entries in the output vector.

Exercise 6.2.

Consider the linear function $G : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ given by

$$G \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad G \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \quad G \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} = G \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = 0.$$

Note that G is already written in some (unknown basis). What is the matrix for G ?

Compute

$$G \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix}.$$

Describe what G does to its inputs in words.

6.2 Maps involving infinite dimensional spaces

We know that some vector spaces do not have a finite dimension (for example, the space of continuous functions $\mathbb{R} \rightarrow \mathbb{R}$). Since the size of a matrix for a linear map is determined by the dimension of the domain (the number of columns) and the dimension of the codomain (the number of rows), it is not possible to represent a linear function to or from an infinite dimensional vector space as a matrix. For now, we don't have any good way to represent these.

6.3 Composing maps

For ordinary functions $\mathbb{R} \rightarrow \mathbb{R}$, we have a way of composing functions. For example, if $f(x) = x^3$ and $g(x) = x^2 + 1$ then the function $f \circ g$ is given by

$$(f \circ g)(x) = f(g(x)) = f(x^2 + 1) = (x^2 + 1)^3.$$

What happens when we compose one linear map with another?

Domains and codomains

First, let us observe that if we want to make a composition $K \circ L$ of two linear functions

$$A \xrightarrow{K} B \qquad C \xrightarrow{L} D$$

it must be true that the outputs of L (its codomain, D) must match the inputs of K (its domain, A).

If we assume that $A = D$, then we can form the composition

$$C \xrightarrow{L} A \xrightarrow{K} B$$

The meaning of $K \circ L$ is exactly the same as in the case of ordinary functions. Given a vector $v \in C$, we can first apply L , get an output, and then apply K to this output:

$$(K \circ L)(v) = K(L(v)).$$

In coordinates

We saw earlier in the chapter that, if $L : C \rightarrow A$ is linear and C, A are finite dimensional, we can represent L as a matrix of numbers (once we have chosen bases for C, A). How do we compose two linear functions once we represent them in coordinates?

Let's compose in the straightforward way. As an example, let R, T be linear functions represented as matrices as

$$R = \begin{bmatrix} 4 & 2 & -1 \\ -2 & 3 & 1 \end{bmatrix}, \quad T = \begin{bmatrix} -1 & 3 & -2 \\ 1 & -2 & 1 \\ 4 & -2 & 3 \end{bmatrix}.$$

Exercise 6.3.

Only one of the two compositions

$$R \circ T, \quad T \circ R$$

can possibly make sense. Which one is it? Explain why the other composition is impossible. (If this is difficult, it will be made clear in the next problem if you didn't choose the correct one.)

Exercise 6.4.

In the previous problem, you concluded only one of the two compositions was possible.

- If you think $R \circ T$ is possible, compute

$$T \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}.$$

Once you have your answer, apply the matrix R to it to get the value of

$$(R \circ T) \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}.$$

- If you think $T \circ R$ is possible, compute

$$R \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}.$$

Once you have your answer, apply the matrix T to it to get the value of

$$(T \circ R) \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}.$$

In the above exercises, you should have concluded that only $R \circ T$ is possible and the result is

$$(R \circ T) \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -28 \\ 34 \end{bmatrix}.$$

Great, now we can compute compositions.

However, it is true that the composition of two linear maps is a linear map itself. This means that $R \circ T$ is a linear function between finite dimensional vector spaces:

$$\begin{array}{ccccc} \mathbb{R}^3 & \xrightarrow{T} & \mathbb{R}^3 & \xrightarrow{R} & \mathbb{R}^2 \\ & \searrow & & \nearrow & \\ & & R \circ T & & \end{array}$$

That is, there should be matrix representation of the function $R \circ T$ that has 2 rows and 3 columns. How can we compute such a matrix?

The algorithm is the same as we used to multiply a matrix by a column.

$$\begin{aligned}
\begin{bmatrix} 4 & 2 & -1 \\ -2 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 & -2 \\ 1 & -2 & 1 \\ 4 & -2 & 3 \end{bmatrix} &= \begin{bmatrix} - & - & - \\ - & - & - \end{bmatrix} \\
&= \begin{bmatrix} 4(-1) + 2(1) + (-1)(4) & - & - \\ - & - & - \end{bmatrix} \\
&= \begin{bmatrix} -6 & - & - \\ - & - & - \end{bmatrix}
\end{aligned}$$

That is, the top left entry in the matrix on the right is the result of combining the **top row** of the left matrix and the **left column** of the right matrix. When we say “combine” here, we mean the same operation we did when multiplying a matrix by a column: the top row and left column are both lists of 3 numbers; we multiply the corresponding entries and add the results.

As another example, let’s compute the bottom middle entry of the result matrix.

$$\begin{aligned}
\begin{bmatrix} 4 & 2 & -1 \\ -2 & 3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 & -2 \\ 1 & -2 & 1 \\ 4 & -2 & 3 \end{bmatrix} &= \begin{bmatrix} -6 & - & - \\ - & (-2)(3) + 3(-2) + 1(-2) & - \end{bmatrix} \\
&= \begin{bmatrix} -6 & - & - \\ - & -14 & - \end{bmatrix}
\end{aligned}$$

Exercise 6.5.

Finish computing the matrix for $R \circ T$.

Exercise 6.6.

Using the matrix for $R \circ T$, compute

$$R \circ T \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

and verify that you get $\begin{bmatrix} -28 \\ 34 \end{bmatrix}$.

Matrix multiplication with `numpy`

Matrix multiplications are long, and humans are prone to make trivial errors. In the wild, you will most often see matrices multiplied using a programming language. As an example, you can compute the matrix for $R \circ T$ using `numpy` for `python` as follows

```
>>> import numpy
>>> R = numpy.array( [ [ 4 , 2 , -1 ] , [ -2 , 3 , 1 ] ] )
>>> T = numpy.array( [ [ -1 , 3 , -2 ] , [ 1 , -2 , 1 ] , [ 4 , -2 , 3 ] ] )
>>> numpy.matmul( R , T )
array([[ -6,  10,  -9],
       [  9, -14,  10]])
```

To get the result

$$R \circ T \begin{bmatrix} 0 \\ -1 \\ 2 \end{bmatrix}$$

you can do (assuming you have already evaluated the previous code block)

```
>>> x = numpy.array( [ [ 0 ] , [ -1 ] , [ 2 ] ] )
>>> RoT = numpy.matmul( R , T )
>>> numpy.matmul( RoT , x )
array([[ -28],
       [  34]])
```

Throughout the remainder of the text, we will show examples of linear algebra computations in a variety of programming languages. (Almost all of these languages use, under the hood, the same two optimized implementations: BLAS and LAPACK.)

Exercise 6.7.

Compute the matrix multiplication

$$\begin{bmatrix} 1 & 3 & 0 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} -2 & -3 & -2 \\ 1 & 2 & -2 \\ 2 & -1 & 3 \end{bmatrix}.$$

Show your work and only (if you want to) use code to check your answer.

Noncommutativity

Beware! Since matrix multiplication corresponds to function composition, and function composition is not commutative, in general matrix multiplication is not guaranteed to be commutative.

That is, if I compute the matrix product

$$AB$$

this will almost always **not** be equal to

$$BA.$$

In fact, in Exercise 6.7 the other matrix product

$$\begin{bmatrix} -1 & 3 & -2 \\ 1 & -2 & 1 \\ 4 & -2 & 3 \end{bmatrix} \begin{bmatrix} 4 & 2 & -1 \\ -2 & 3 & 1 \end{bmatrix}$$

is not even defined. (Why?)

It is, however, true that matrix multiplication is associative. That is, if I want to multiply three matrices

$$ABC = A(BC) = (AB)C$$

I can multiply the right two first to get one matrix representing BC , then multiply on the left by A to get one matrix representing ABC .

I could also multiply starting from the left and obtain the same answer.

All three of the ways of multiplying the matrices above results in a single matrix representing the composition

$$\begin{array}{ccccccc} \mathbb{R}^k & \xrightarrow{C} & \mathbb{R}^\ell & \xrightarrow{B} & \mathbb{R}^m & \xrightarrow{A} & \mathbb{R}^n \\ & \searrow & & & \nearrow & & \\ & & A \circ B \circ C & & & & \end{array}$$

Figure 6.1

6.4 Algebra of linear functions

Note that when we have two functions $\mathbb{R} \rightarrow \mathbb{R}$, we can add them and multiply by constants. For example, consider $f(x) = e^x$ and $g(x) = x^2 + 1$. We can form the new function

$$f(x) + 2g(x)$$

which just means compute f on the input x , then compute g on the input x and multiply the result by 2, then add the two parts you got. So

$$f(x) + 2g(x) = e^x + 2(x^2 + 1).$$

What is really happening is that we are evaluating f and g separately on x . Since they have the same codomain, once we have $f(x), g(x)$ which are objects in their codomain, we can do algebra in the codomain as usual.

The same is true of linear maps and so, in finite dimensions with fixed bases, must be true of matrices as well. There is one complication, however: in order to combine linear functions (and the matrices representing them) in this way, we must first check that they have the same codomain.

As an example, consider $F, G : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ given by

$$F = \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 4 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 0 \\ -1 & 1 \\ 2 & 0 \end{bmatrix}.$$

We can compute the matrix for $F + 3G$ in the following way

$$\begin{aligned} \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 4 & 1 \end{bmatrix} + 3 \begin{bmatrix} 0 & 0 \\ -1 & 1 \\ 2 & 0 \end{bmatrix} &= \begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 4 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -3 & 3 \\ 6 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ -2 & 2 \\ 10 & 1 \end{bmatrix} \end{aligned}$$

That is, to multiply a linear map (written as a matrix) by a constant k , we multiply all entries of the matrix by k . To add linear maps (written as matrices), we add corresponding entries.

Note that the addition

$$\begin{bmatrix} 1 & 2 \\ 1 & -1 \\ 4 & 1 \end{bmatrix} + \begin{bmatrix} -1 & 1 \\ 2 & 0 \end{bmatrix}$$

is not defined since the domain and codomain of these matrices have different dimensions.

Exercise 6.8.

What are the dimensions of the domain and codomain of the two matrices above?

Exercise 6.9.

Compute

$$\frac{1}{2} \begin{bmatrix} 0 & 1 & 0 & -2 \\ 1 & 3 & 0 & -1 \\ 0 & 2 & -3 & -1 \\ 0 & 1 & 1 & 0 \end{bmatrix} + 2 \begin{bmatrix} 0 & 3 & -2 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & -1 & -1 \\ 0 & 2 & 1 & 1 \end{bmatrix}.$$

Chapter 7

The usefulness of matrix equations and their solutions

In the previous chapter, we saw that we can use a matrix to represent (in coordinates) a linear function, and we saw how to compute the output (in coordinates) when we apply a linear function to a vector (in coordinates).

We also saw that we can represent the composition of two linear functions as a single matrix by multiplying their matrices.

7.1 Matrix multiplication and linear combinations

Recall that the method to find a matrix for a linear function $F : A \rightarrow B$ is the following:

1. Choose a basis $[b_1, b_2, \dots, b_k]$ for A .
2. Choose a basis $[c_1, c_2, \dots, c_\ell]$ for B .
3. Compute the coefficients of $F(b_1) \in B$ with respect to $[c_1, c_2, \dots, c_\ell]$.
4. Take that list of coefficients and write it as the first column of our result matrix:

$$[F(b_1) \mid \cdots]$$

5. Repeat steps (3.) and (4.) for the output of F on the second basis element b_2 . This is the second column of the matrix for F :

$$[F(b_1) \mid F(b_2) \mid \cdots]$$

6. Continue until there are no basis elements in the domain left.

We already saw the example

$$F = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

at the beginning of Section 6.1.

What we noticed then is that the multiplication

$$F \begin{bmatrix} a \\ b \\ c \end{bmatrix} = a \begin{bmatrix} 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

really just yields an arbitrary linear combination of the columns of F .

This holds in general: if A is a matrix with columns $A_1, A_2, \dots, A_n$ and X is the vector

$$x := \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Then the multiplication of A and x is

$$Ax = x_1 A_1 + x_2 A_2 + \cdots + x_n A_n.$$

That is, the multiplication Ax is a linear combination of the columns of A . The coefficients of the linear combination are the entries in the column vector x .

7.2 Very many linear combinations

Now, recall that many concepts we have seen thus far involve linear combinations. For example:

- Definition 3.3 can be rephrased as:

The span of the columns of A is exactly the set

$$\{Ax \mid \forall x\}.$$

- Theorem 3.3 has, as a consequence the following:

The columns of A are linearly independent if and only if the matrix equation

$$Ax = 0$$

has only one solution ($x = 0$).

- Definition 3.5 can be rephrased as:

The columns of A form a basis for its codomain if and only if the following two properties hold:

1. For each y in the codomain of A , the equation $Ax = y$ has a solution (that is, A is surjective).
2. $Ax = 0$ has only one solution.

Many more statements so far can be rephrased in terms of matrix equations.

Exercise 7.1.

Rewrite the following system of equations as a matrix equation:

$$\begin{aligned} 3x + 2y &= 5 \\ -x + y &= 1. \end{aligned}$$

Exercise 7.2.

Rewrite the following system of equations as a matrix equation:

$$\begin{aligned} x - 2y &= 3 \\ y &= 0 \\ -z + y - 1 &= 0. \end{aligned}$$

Chapter 8

Inverses of linear maps and matrices

[TODO]

8.1 Identities and inverses

Let's go back and talk about functions $\mathbb{R} \rightarrow \mathbb{R}$ again. One thing we have for such functions is an identity function

$$I(x) = x.$$

We give this the name “identity” and use the symbol I for it, but what in practice does this mean? What are the properties an identity should have?

Definition 8.1. Given a collection S of things and an operation $\cdot : S \times S \rightarrow S$ for combining two things in S , a *left identity* for $\cdot$ is an element $e \in S$ that satisfies

$$e \cdot p = p, \quad \forall p \in S.$$

A *right identity* for $\cdot$ is an element $\hat{e} \in S$ such that

$$p \cdot \hat{e} = p, \quad \forall p \in S.$$

That is, doing an operation with its left identity on the left side does nothing. Similarly for the right side.

We have already seen some identity objects so far:

- 0 is a left (and right) identity for $+$ in $\mathbb{R}$ (or by definition any group, field, etc)

- 1 is a left (and right) identity for multiplication in $\mathbb{R}$ (or by definition any field, etc)
- the zero vector in a vector space is (by definition) a left (and right) identity for vector addition in that space

The function $I(x) = x$ is a left (and right) identity for a different operation in $\mathbb{R} \rightarrow \mathbb{R}$; it is the identity for composition. That is, for any $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$I \circ f = f \quad \text{and} \quad f \circ I = f.$$

In more detail, this looks like

$$(I \circ f)(x) = I(f(x)) = f(x) = f(I(x)) = (f \circ I)(x).$$

If we have an operation and an identity for that operation, we may also be lucky enough to have inverses for that operation:

Definition 8.2. Given a collection S of things and an operation $\cdot : S \times S \rightarrow S$ for combining two things in S :

- if $\cdot$ has a left identity e_L , then an element $p \in S$ has a *left inverse* if there is an element $k \in S$ such that

$$k \cdot p = e_L.$$

- if $\cdot$ has a right identity e_R , then an element $p \in S$ has a *right inverse* if there is an element $\hat{k} \in S$ such that

$$p \cdot \hat{k} = e_R.$$

All the operations we have considered so far have the same left and right inverse, and are associative. In this case, one can show¹ that the left and right inverses, if they exist, are the same. In this case, we simply call it “the inverse” of (eg) x and write it as x^{-1} .

As an example, consider the function $f(x) = x^3 + 1$. One can check that $g(x) = (x - 1)^{1/3}$ is both a left and right inverse for composition, and so we would normally write $f^{-1}(x) = (x - 1)^{1/3}$.

We have seen other inverse elements:

- Given $a \in \mathbb{R}$, there is an element (which we usually write $-a$) which is the inverse of a with respect to the operation $+$. 0 is its own inverse w.r.t. $+$.
- Given $a \in \mathbb{R}$ which is not zero, there is an element (which we usually write $\frac{1}{a}$) which is the inverse of a with respect to the operation $\cdot$. 1 is its own inverse w.r.t. $\cdot$.
- Given a vector $v \in V$ in a vector space, there is (by definition) a vector $-v \in V$ which is an inverse for vector addition.

¹One usually learns to prove this in a course on Abstract Algebra. I have mentioned in an earlier footnote a source for this.

8.2 The identity for linear functions

If we have a linear function $G : A \rightarrow B$, what is the identity for composition? Well, if there is a left identity I_L , it must satisfy $I_L \circ G = G$. Let's write this out in a diagram:

$$\begin{array}{ccccc} A & \xrightarrow{G} & B & \xrightarrow{I_L} & ? \\ & \searrow & & \nearrow & \\ & & I_L \circ G = G & & \end{array}$$

Since the map on the bottom of the diagram is G , we know that the $?$ on the right must be the codomain of G , that is, the set B :

$$\begin{array}{ccccc} A & \xrightarrow{G} & B & \xrightarrow{I_L} & B \\ & \searrow & & \nearrow & \\ & & I_L \circ G = G & & \end{array}$$

So, $I_L : B \rightarrow B$ is a linear function which doesn't change its inputs. It's straightforward to guess that $I_L(v) = v$.

What about a right inverse I_R ? By the same reasoning,

$$\begin{array}{ccccc} A & \xrightarrow{I_R} & A & \xrightarrow{G} & B \\ & \searrow & & \nearrow & \\ & & G \circ I_R = G & & \end{array}$$

It is straightforward to guess that $I_R(v) = v$ as well.

But beware! The left and right inverses are not the same function. I_L has domain and codomain B while I_R has domain and codomain A .

However, these two inverses have something in common. In coordinates, they both have the form

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

the only potential difference is the size.

Definition 8.3. For every vector space V there is an identity function, which is linear, from V to V . In general, it takes its input and returns them unmodified:

$$I(v) = v.$$

If V is finite dimensional and we fix a basis for it, then $I : F^k \rightarrow F^k$.

Lemma 8.1. *In the latter case of Definition 8.3, in coordinates I has the form*

$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

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